Table 1: Variational vs classical · mixture cloud ($N = 128$, $d = 2$): sliced W1. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	sliced W1
1	particles	**ssp**	100/100	$\mathbf{0.1208 \pm 0.01435}$
2	particles	multinomial	100/100	0.1953 ± 0.01885
3	variational	kl	100/100	0.652 ± 0
3	variational	tv	100/100	0.652 ± 0
5	particles	systematic	100/100	0.8868 ± 0.07697

Table 2: Variational vs classical · mixture cloud ($N = 128$, $d = 2$): trials keeping both modes. Best = higher is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	trials keeping both modes
1	variational	**kl**	100/100	$\mathbf{1 \pm 0}$
1	variational	**tv**	100/100	$\mathbf{1 \pm 0}$
1	particles	**ssp**	100/100	$\mathbf{1 \pm 0}$
1	particles	**multinomial**	100/100	$\mathbf{1 \pm 0}$
5	particles	systematic	100/100	0.865 ± 0.04427

Table 3: Variational vs classical · mixture cloud ($N = 128$, $d = 2$): forward time (ms). Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	forward time (ms)
1	particles	**ssp**	100/100	$\mathbf{0.1907 \pm 0.005689}$
2	particles	systematic	100/100	0.2093 ± 0.009762
3	particles	multinomial	100/100	0.3192 ± 0.02899
4	variational	tv	100/100	1.109 ± 0.1258
5	variational	kl	100/100	2.227 ± 0.1671

Table 4: Variational vs classical · Linear Gaussian filter ($N = 128$): RMSE. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	RMSE
1	particles	**systematic**	100/100	$\mathbf{0.5061 \pm 0.04637}$
2	particles	multinomial	100/100	0.5065 ± 0.04512
3	particles	ssp	100/100	0.5068 ± 0.04587
4	variational	kl	100/100	0.507 ± 0.04685
5	variational	tv	100/100	0.5081 ± 0.04693

Table 5: Variational vs classical · UNGM filter ($N = 128$): RMSE. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	RMSE
1	variational	**kl**	100/100	$\mathbf{4.831 \pm 0.8575}$
2	particles	ssp	100/100	4.854 ± 0.792
3	particles	systematic	100/100	4.906 ± 0.9289
4	variational	tv	100/100	4.906 ± 0.794
5	particles	multinomial	100/100	4.916 ± 0.8803

Table 6: Variational vs classical · Stochastic volatility filter ($N = 128$): RMSE. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	RMSE
1	particles	**ssp**	100/100	$\mathbf{0.4164 \pm 0.06348}$
2	particles	systematic	100/100	0.4173 ± 0.0634
3	particles	multinomial	100/100	0.4174 ± 0.06486
4	variational	kl	100/100	0.4174 ± 0.06379
5	variational	tv	100/100	0.4181 ± 0.06438

Table 7: Variational vs classical · Bearing-only tracking filter ($N = 128$): RMSE. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	RMSE
1	particles	**systematic**	100/100	$\mathbf{4.548 \pm 1.657}$
2	particles	ssp	100/100	4.649 ± 1.688
3	variational	kl	100/100	4.662 ± 1.626
4	particles	multinomial	100/100	4.787 ± 1.635
5	variational	tv	100/100	4.869 ± 1.788

Table 8: Variational vs classical · Linear Gaussian filter ($N = 128$): conditional log-likelihood variance. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	conditional log-likelihood variance
1	variational	**tv**	100/100	**0.3037**
2	particles	ssp	100/100	0.3507
3	particles	systematic	100/100	0.3526
4	variational	kl	100/100	0.358
5	particles	multinomial	100/100	0.4062

Table 9: Variational vs classical · UNGM filter ($N = 128$): conditional log-likelihood variance. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	conditional log-likelihood variance
1	particles	**ssp**	100/100	**633.7**
2	particles	systematic	100/100	653.3
3	variational	kl	100/100	756
4	variational	tv	100/100	2923
5	particles	multinomial	100/100	4174

Table 10: Variational vs classical · Stochastic volatility filter ($N = 128$): conditional log-likelihood variance. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	conditional log-likelihood variance
1	particles	**systematic**	100/100	**0.08285**
2	particles	ssp	100/100	0.08356
3	variational	tv	100/100	0.08647
4	variational	kl	100/100	0.09313
5	particles	multinomial	100/100	0.107

Table 11: Variational vs classical · Bearing-only tracking filter ($N = 128$): conditional log-likelihood variance. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	conditional log-likelihood variance
1	particles	**ssp**	100/100	**2.019**
2	particles	systematic	100/100	4.739
3	variational	tv	100/100	6.066
4	variational	kl	100/100	7.522
5	particles	multinomial	100/100	19.1

Table 12: Mixture cloud ($N = 128$, $d = 2$): sliced W1. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	sliced W1
1	stonesoup	**systematic**	100/100	$\mathbf{0.04703 \pm 0.001707}$
2	julia	systematic	100/100	0.05308 ± 0.001836
3	stonesoup	stratified	100/100	0.05856 ± 0.002222
4	julia	stratified	100/100	0.1159 ± 0.01168
5	filterflow	stratified	100/100	0.1171 ± 0.01285
6	particles	ssp	100/100	0.1208 ± 0.01435
7	julia	killing	100/100	0.1209 ± 0.01291
8	particles	stratified	100/100	0.1236 ± 0.01296
9	particles	residual	100/100	0.1285 ± 0.01141
10	julia	ssp	100/100	0.1303 ± 0.01494
11	julia	residual	100/100	0.1364 ± 0.0113
12	stonesoup	residual	100/100	0.1405 ± 0.01307
13	cuthbert	multinomial	100/100	0.1669 ± 0.01554
14	julia	multinomial	100/100	0.1841 ± 0.01849
15	pydpf	soft	100/100	0.186 ± 0.01745
15	torchfilter	soft	100/100	0.186 ± 0.01745
17	pfjax	multinomial	100/100	0.1892 ± 0.01716
18	pydpf	multinomial	100/100	0.1935 ± 0.01816
18	torchfilter	multinomial	100/100	0.1935 ± 0.01816
20	particles	multinomial	100/100	0.1953 ± 0.01885
21	filterflow	multinomial	100/100	0.2042 ± 0.01813
22	stonesoup	multinomial	100/100	0.2066 ± 0.02016
23	pfjax	ot	10/10	0.4299 ± 0
24	pydpf	ot	10/10	0.4356 ± 0
25	filterflow	corrected_ot	10/10	0.6283 ± 0
25	filterflow	ot	10/10	0.6283 ± 0
27	variational	kl	100/100	0.652 ± 0
27	variational	tv	100/100	0.652 ± 0
29	custom	eigen	100/100	0.7163 ± 0
30	custom	gumbel	10/10	0.7719 ± 0.038
31	pfjax	mvn	100/100	0.8582 ± 0.01277
32	custom	gaussian	100/100	0.8599 ± 0.0134
33	particles	systematic	100/100	0.8868 ± 0.07697
34	filterflow	systematic	100/100	0.9303 ± 0.08088
35	pydpf	systematic	100/100	0.9303 ± 0.08088
35	pydpf	systematic_stop	100/100	0.9303 ± 0.08088
35	cuthbert	systematic	100/100	0.9303 ± 0.08088
35	cuthbert	systematic_stop	100/100	0.9303 ± 0.08088

Table 13: Mixture cloud ($N = 128$, $d = 2$): RFF MMD^2. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	RFF MMD^2
1	stonesoup	**systematic**	100/100	$\mathbf{0.0001244 \pm 1.445 \times 10^{-5}}$
2	julia	systematic	100/100	$0.000196 \pm 2.053 \times 10^{-5}$
3	stonesoup	stratified	100/100	$0.000223 \pm 2.897 \times 10^{-5}$
4	julia	stratified	100/100	0.001388 ± 0.0003045
5	filterflow	stratified	100/100	0.001477 ± 0.0003767
6	julia	killing	100/100	0.00148 ± 0.0003833
7	particles	residual	100/100	0.001523 ± 0.0003433
8	particles	ssp	100/100	0.001649 ± 0.0005011
9	particles	stratified	100/100	0.001665 ± 0.0003661
10	julia	residual	100/100	0.001711 ± 0.0003182
11	stonesoup	residual	100/100	0.001896 ± 0.0003947
12	julia	ssp	100/100	0.001898 ± 0.0004742
13	cuthbert	multinomial	100/100	0.00263 ± 0.000644
14	pydpf	soft	100/100	0.003264 ± 0.0007189
14	torchfilter	soft	100/100	0.003264 ± 0.0007189
16	julia	multinomial	100/100	0.003346 ± 0.0007996
17	pfjax	multinomial	100/100	0.003463 ± 0.0007332
18	pydpf	multinomial	100/100	0.003563 ± 0.0007708
18	torchfilter	multinomial	100/100	0.003563 ± 0.0007708
20	particles	multinomial	100/100	0.003629 ± 0.0007993
21	filterflow	multinomial	100/100	0.003941 ± 0.0007707
22	stonesoup	multinomial	100/100	0.004211 ± 0.0008681
23	pydpf	ot	10/10	0.01226 ± 0
24	pfjax	ot	10/10	0.0126 ± 0
25	custom	eigen	100/100	0.02422 ± 0
26	filterflow	corrected_ot	10/10	0.02492 ± 0
26	filterflow	ot	10/10	0.02492 ± 0
28	pfjax	mvn	100/100	0.0315 ± 0.002173
29	custom	gaussian	100/100	0.0325 ± 0.001953
30	variational	kl	100/100	0.03978 ± 0
30	variational	tv	100/100	0.03978 ± 0
32	custom	gumbel	10/10	0.05546 ± 0.00518
33	particles	systematic	100/100	0.08753 ± 0.01565
34	filterflow	systematic	100/100	0.09637 ± 0.01645
35	pydpf	systematic	100/100	0.09637 ± 0.01645
35	pydpf	systematic_stop	100/100	0.09637 ± 0.01645
35	cuthbert	systematic	100/100	0.09637 ± 0.01645
35	cuthbert	systematic_stop	100/100	0.09637 ± 0.01645

Table 14: Mixture cloud ($N = 128$, $d = 2$): covariance error (Frobenius). Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	covariance error (Frobenius)
1	custom	**eigen**	100/100	$\mathbf{1.311 \times 10^{-15} \pm 0}$
2	stonesoup	systematic	100/100	0.1629 ± 0.01208
3	julia	systematic	100/100	0.2301 ± 0.01716
4	stonesoup	stratified	100/100	0.2409 ± 0.02174
5	filterflow	stratified	100/100	0.3263 ± 0.03138
6	julia	stratified	100/100	0.3264 ± 0.03411
7	particles	stratified	100/100	0.3337 ± 0.04024
8	particles	ssp	100/100	0.3442 ± 0.04689
9	julia	killing	100/100	0.3478 ± 0.04049
10	julia	ssp	100/100	0.3662 ± 0.04016
11	particles	residual	100/100	0.405 ± 0.03817
12	julia	residual	100/100	0.4303 ± 0.04155
13	cuthbert	multinomial	100/100	0.4797 ± 0.0496
14	stonesoup	residual	100/100	0.4878 ± 0.04743
15	pfjax	multinomial	100/100	0.5114 ± 0.05508
16	pydpf	soft	100/100	0.5216 ± 0.05562
16	torchfilter	soft	100/100	0.5216 ± 0.05562
18	julia	multinomial	100/100	0.5409 ± 0.05384
19	particles	multinomial	100/100	0.5588 ± 0.058
20	pydpf	multinomial	100/100	0.5698 ± 0.06163
20	torchfilter	multinomial	100/100	0.5698 ± 0.06163
22	stonesoup	multinomial	100/100	0.5729 ± 0.06508
23	filterflow	multinomial	100/100	0.5978 ± 0.0549
24	variational	kl	100/100	0.833 ± 0
24	variational	tv	100/100	0.833 ± 0
26	pfjax	mvn	100/100	0.9077 ± 0.1145
27	custom	gaussian	100/100	0.9295 ± 0.1319
28	pydpf	ot	10/10	1.925 ± 0
29	pfjax	ot	10/10	2.134 ± 0
30	filterflow	corrected_ot	10/10	2.857 ± 0
30	filterflow	ot	10/10	2.857 ± 0
32	particles	systematic	100/100	2.906 ± 0.6798
33	filterflow	systematic	100/100	3.29 ± 0.7143
34	pydpf	systematic	100/100	3.29 ± 0.7143
34	pydpf	systematic_stop	100/100	3.29 ± 0.7143
34	cuthbert	systematic	100/100	3.29 ± 0.7143
34	cuthbert	systematic_stop	100/100	3.29 ± 0.7143
38	custom	gumbel	10/10	4.487 ± 0.3302

Table 15: Mixture cloud ($N = 128$, $d = 2$): trials keeping both modes. Best = higher is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	trials keeping both modes
1	custom	**eigen**	100/100	$\mathbf{1 \pm 0}$
1	custom	**gaussian**	100/100	$\mathbf{1 \pm 0}$
1	custom	**gumbel**	10/10	$\mathbf{1 \pm 0}$
1	cuthbert	**multinomial**	100/100	$\mathbf{1 \pm 0}$
1	filterflow	**corrected_ot**	10/10	$\mathbf{1 \pm 0}$
1	filterflow	**multinomial**	100/100	$\mathbf{1 \pm 0}$
1	filterflow	**ot**	10/10	$\mathbf{1 \pm 0}$
1	filterflow	**stratified**	100/100	$\mathbf{1 \pm 0}$
1	julia	**killing**	100/100	$\mathbf{1 \pm 0}$
1	julia	**multinomial**	100/100	$\mathbf{1 \pm 0}$
1	julia	**residual**	100/100	$\mathbf{1 \pm 0}$
1	julia	**ssp**	100/100	$\mathbf{1 \pm 0}$
1	julia	**stratified**	100/100	$\mathbf{1 \pm 0}$
1	julia	**systematic**	100/100	$\mathbf{1 \pm 0}$
1	particles	**multinomial**	100/100	$\mathbf{1 \pm 0}$
1	particles	**residual**	100/100	$\mathbf{1 \pm 0}$
1	particles	**ssp**	100/100	$\mathbf{1 \pm 0}$
1	particles	**stratified**	100/100	$\mathbf{1 \pm 0}$
1	pfjax	**multinomial**	100/100	$\mathbf{1 \pm 0}$
1	pfjax	**mvn**	100/100	$\mathbf{1 \pm 0}$
1	pfjax	**ot**	10/10	$\mathbf{1 \pm 0}$
1	pydpf	**multinomial**	100/100	$\mathbf{1 \pm 0}$
1	pydpf	**ot**	10/10	$\mathbf{1 \pm 0}$
1	pydpf	**soft**	100/100	$\mathbf{1 \pm 0}$
1	stonesoup	**multinomial**	100/100	$\mathbf{1 \pm 0}$
1	stonesoup	**residual**	100/100	$\mathbf{1 \pm 0}$
1	stonesoup	**stratified**	100/100	$\mathbf{1 \pm 0}$
1	stonesoup	**systematic**	100/100	$\mathbf{1 \pm 0}$
1	torchfilter	**multinomial**	100/100	$\mathbf{1 \pm 0}$
1	torchfilter	**soft**	100/100	$\mathbf{1 \pm 0}$
1	variational	**kl**	100/100	$\mathbf{1 \pm 0}$
1	variational	**tv**	100/100	$\mathbf{1 \pm 0}$
33	particles	systematic	100/100	0.865 ± 0.04427
34	cuthbert	systematic	100/100	0.84 ± 0.04651
34	cuthbert	systematic_stop	100/100	0.84 ± 0.04651
34	filterflow	systematic	100/100	0.84 ± 0.04651
34	pydpf	systematic	100/100	0.84 ± 0.04651
34	pydpf	systematic_stop	100/100	0.84 ± 0.04651

Table 16: Mixture cloud ($N = 128$, $d = 2$): forward time (ms). Best = lower is better. Best method in bold; $\pm$ gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	forward time (ms)
1	particles	**ssp**	100/100	$\mathbf{0.1907 \pm 0.005689}$
2	particles	systematic	100/100	0.2093 ± 0.009762
3	particles	stratified	100/100	0.2263 ± 0.01447
4	particles	residual	100/100	0.2528 ± 0.01066
5	particles	multinomial	100/100	0.3192 ± 0.02899
6	julia	stratified	100/100	0.343 ± 0.01239
7	julia	ssp	100/100	0.3633 ± 0.01138
8	julia	multinomial	100/100	0.3702 ± 0.01667
9	julia	residual	100/100	0.3709 ± 0.01241
10	julia	systematic	100/100	0.3779 ± 0.01573
11	stonesoup	systematic	100/100	0.3949 ± 0.01314
12	stonesoup	stratified	100/100	0.4062 ± 0.01413
13	julia	killing	100/100	0.4145 ± 0.01357
14	stonesoup	multinomial	100/100	0.415 ± 0.02028
15	stonesoup	residual	100/100	0.5988 ± 0.0498
16	pydpf	multinomial	100/100	0.6702 ± 0.02421
17	custom	gaussian	100/100	0.7702 ± 0.02496
18	pydpf	soft	100/100	0.8031 ± 0.03183
19	torchfilter	multinomial	100/100	0.8181 ± 0.02647
20	pydpf	systematic	100/100	0.8183 ± 0.03524
21	pydpf	systematic_stop	100/100	0.827 ± 0.02605
22	pfjax	multinomial	100/100	0.9381 ± 0.0229
23	filterflow	multinomial	100/100	0.947 ± 0.02772
24	filterflow	stratified	100/100	0.9598 ± 0.03241
25	filterflow	systematic	100/100	0.9602 ± 0.03042
26	custom	gumbel	10/10	0.9708 ± 0.08369
27	custom	eigen	100/100	1.013 ± 0.06631
28	pfjax	mvn	100/100	1.027 ± 0.02351
29	variational	tv	100/100	1.109 ± 0.1258
30	torchfilter	soft	100/100	1.152 ± 0.03697
31	cuthbert	systematic_stop	100/100	1.362 ± 0.03292
32	cuthbert	systematic	100/100	1.391 ± 0.04303
33	cuthbert	multinomial	100/100	1.926 ± 0.05821
34	variational	kl	100/100	2.227 ± 0.1671
35	filterflow	ot	10/10	11.67 ± 0.5836
36	filterflow	corrected_ot	10/10	26.17 ± 1.347
37	pfjax	ot	10/10	27.64 ± 2.206
38	pydpf	ot	10/10	33.95 ± 2.185

Table 17: Linear Gaussian filter ($N = 128$): RMSE. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	RMSE
1	julia	**multinomial**	100/100	$\mathbf{0.505 \pm 0.04559}$
2	particles	residual	100/100	0.5054 ± 0.04552
3	stonesoup	residual	100/100	0.5056 ± 0.0462
4	pfjax	multinomial	100/100	0.5056 ± 0.04498
5	pfjax	mvn	100/100	0.5058 ± 0.04612
6	pydpf	multinomial	100/100	0.5058 ± 0.04658
6	torchfilter	multinomial	100/100	0.5058 ± 0.04658
8	stonesoup	multinomial	100/100	0.5058 ± 0.04639
9	cuthbert	multinomial	100/100	0.5058 ± 0.04598
10	filterflow	multinomial	100/100	0.506 ± 0.0471
11	particles	systematic	100/100	0.5061 ± 0.04637
12	julia	residual	100/100	0.5061 ± 0.0458
13	pydpf	systematic	100/100	0.5064 ± 0.0469
13	pydpf	systematic_stop	100/100	0.5064 ± 0.0469
15	stonesoup	systematic	100/100	0.5064 ± 0.04698
16	cuthbert	systematic	100/100	0.5065 ± 0.04593
16	cuthbert	systematic_stop	100/100	0.5065 ± 0.04593
18	particles	multinomial	100/100	0.5065 ± 0.04512
19	julia	stratified	100/100	0.5066 ± 0.04604
20	julia	killing	100/100	0.5066 ± 0.04645
21	particles	stratified	100/100	0.5066 ± 0.04543
22	filterflow	stratified	100/100	0.5066 ± 0.04594
23	filterflow	systematic	100/100	0.5067 ± 0.04571
24	julia	systematic	100/100	0.5067 ± 0.04616
25	particles	ssp	100/100	0.5068 ± 0.04587
26	custom	eigen	100/100	0.5069 ± 0.0455
27	julia	ssp	100/100	0.507 ± 0.04575
28	variational	kl	100/100	0.507 ± 0.04685
29	stonesoup	stratified	100/100	0.5072 ± 0.04628
30	custom	gaussian	100/100	0.5072 ± 0.04556
31	variational	tv	100/100	0.5081 ± 0.04693
32	pydpf	soft	100/100	0.5082 ± 0.04603
32	torchfilter	soft	100/100	0.5082 ± 0.04603
34	filterflow	corrected_ot	10/10	0.5144 ± 0.04365
35	custom	gumbel	10/10	0.5144 ± 0.04802
36	filterflow	ot	10/10	0.5145 ± 0.04393
37	pydpf	ot	10/10	0.5145 ± 0.04344
38	pfjax	ot	10/10	0.5146 ± 0.04344

Table 18: Linear Gaussian filter ($N = 128$): 95% coverage. Best = closest to 95% is better. Best method in bold; $\pm$ gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	95% coverage
1	filterflow	**corrected_ot**	10/10	$\mathbf{0.932 \pm 0.02154}$
2	stonesoup	residual	100/100	0.9316 ± 0.02163
3	julia	residual	100/100	0.9308 ± 0.02094
3	pydpf	systematic	100/100	0.9308 ± 0.02122
3	pydpf	systematic_stop	100/100	0.9308 ± 0.02122
6	particles	stratified	100/100	0.9306 ± 0.02209
6	pydpf	multinomial	100/100	0.9306 ± 0.02266
6	torchfilter	multinomial	100/100	0.9306 ± 0.02266
9	cuthbert	multinomial	100/100	0.9304 ± 0.01987
9	particles	residual	100/100	0.9304 ± 0.01918
11	pfjax	ot	10/10	0.93 ± 0.02051
12	julia	killing	100/100	0.9296 ± 0.02241
12	julia	stratified	100/100	0.9296 ± 0.0213
14	julia	ssp	100/100	0.9294 ± 0.02137
14	variational	kl	100/100	0.9294 ± 0.01994
16	custom	gaussian	100/100	0.9292 ± 0.02216
16	particles	systematic	100/100	0.9292 ± 0.02184
18	particles	ssp	100/100	0.929 ± 0.02118
19	stonesoup	multinomial	100/100	0.9288 ± 0.0217
19	custom	eigen	100/100	0.9288 ± 0.02219
21	filterflow	multinomial	100/100	0.9286 ± 0.02167
21	pfjax	multinomial	100/100	0.9286 ± 0.02243
21	pfjax	mvn	100/100	0.9286 ± 0.02183
24	julia	multinomial	100/100	0.9284 ± 0.0214
24	variational	tv	100/100	0.9284 ± 0.02214
26	custom	gumbel	10/10	0.928 ± 0.02154
26	pydpf	ot	10/10	0.928 ± 0.02154
28	filterflow	stratified	100/100	0.9278 ± 0.02192
28	stonesoup	stratified	100/100	0.9278 ± 0.02262
30	cuthbert	systematic	100/100	0.9276 ± 0.02216
30	cuthbert	systematic_stop	100/100	0.9276 ± 0.02216
30	particles	multinomial	100/100	0.9276 ± 0.02232
33	filterflow	systematic	100/100	0.9274 ± 0.0219
33	pydpf	soft	100/100	0.9274 ± 0.0228
33	torchfilter	soft	100/100	0.9274 ± 0.0228
36	julia	systematic	100/100	0.9264 ± 0.02278
37	stonesoup	systematic	100/100	0.9262 ± 0.02292
38	filterflow	ot	10/10	0.924 ± 0.02001

Table 19: Linear Gaussian filter ($N = 128$): conditional log-likelihood variance. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	conditional log-likelihood variance
1	julia	**killing**	100/100	**0.2993**
2	variational	tv	100/100	0.3037
3	stonesoup	systematic	100/100	0.3047
4	custom	eigen	100/100	0.3199
5	cuthbert	systematic	100/100	0.321
5	cuthbert	systematic_stop	100/100	0.321
7	stonesoup	stratified	100/100	0.3259
8	pydpf	systematic	100/100	0.327
8	pydpf	systematic_stop	100/100	0.327
10	particles	stratified	100/100	0.348
11	julia	ssp	100/100	0.3498
12	particles	ssp	100/100	0.3507
13	filterflow	stratified	100/100	0.3512
14	particles	systematic	100/100	0.3526
15	variational	kl	100/100	0.358
16	filterflow	multinomial	100/100	0.3587
17	stonesoup	multinomial	100/100	0.3589
18	pfjax	multinomial	100/100	0.3605
19	stonesoup	residual	100/100	0.3605
20	julia	systematic	100/100	0.3609
21	filterflow	systematic	100/100	0.3659
22	particles	residual	100/100	0.3669
23	pydpf	multinomial	100/100	0.374
23	torchfilter	multinomial	100/100	0.374
25	cuthbert	multinomial	100/100	0.3784
26	julia	multinomial	100/100	0.3786
27	particles	multinomial	100/100	0.4062
28	pfjax	mvn	100/100	0.4128
29	julia	stratified	100/100	0.4132
30	custom	gaussian	100/100	0.4173
31	pydpf	soft	100/100	0.4185
31	torchfilter	soft	100/100	0.4185
33	julia	residual	100/100	0.435
–	custom	gumbel	10/10	–
–	filterflow	corrected_ot	10/10	–
–	filterflow	ot	10/10	–
–	pfjax	ot	10/10	–
–	pydpf	ot	10/10	–

Table 20: UNGM filter ($N = 128$): RMSE. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	RMSE
1	julia	**killing**	100/100	$\mathbf{4.762 \pm 0.8073}$
2	filterflow	multinomial	100/100	4.79 ± 0.8229
3	filterflow	systematic	100/100	4.792 ± 0.8378
4	particles	stratified	100/100	4.824 ± 0.8784
5	pydpf	soft	100/100	4.825 ± 0.7488
5	torchfilter	soft	100/100	4.825 ± 0.7488
7	pydpf	multinomial	100/100	4.827 ± 0.8482
7	torchfilter	multinomial	100/100	4.827 ± 0.8482
9	variational	kl	100/100	4.831 ± 0.8575
10	stonesoup	multinomial	100/100	4.843 ± 0.8865
11	filterflow	stratified	100/100	4.853 ± 0.8544
12	particles	ssp	100/100	4.854 ± 0.792
13	julia	ssp	100/100	4.86 ± 0.9073
14	cuthbert	systematic	100/100	4.87 ± 0.9592
14	cuthbert	systematic_stop	100/100	4.87 ± 0.9592
16	julia	multinomial	100/100	4.871 ± 0.7691
17	pydpf	systematic	100/100	4.881 ± 0.8893
17	pydpf	systematic_stop	100/100	4.881 ± 0.8893
19	julia	residual	100/100	4.888 ± 0.9887
20	particles	systematic	100/100	4.906 ± 0.9289
21	variational	tv	100/100	4.906 ± 0.794
22	julia	stratified	100/100	4.906 ± 0.9287
23	particles	multinomial	100/100	4.916 ± 0.8803
24	julia	systematic	100/100	4.921 ± 0.9203
25	stonesoup	residual	100/100	4.935 ± 0.9515
26	stonesoup	systematic	100/100	4.935 ± 0.8928
27	custom	gaussian	100/100	4.936 ± 0.8551
28	particles	residual	100/100	4.942 ± 1.015
29	cuthbert	multinomial	100/100	4.972 ± 0.9957
30	pfjax	mvn	93/100	5.027 ± 0.9052
31	pfjax	multinomial	100/100	5.038 ± 0.8834
32	stonesoup	stratified	100/100	5.039 ± 0.9313
33	custom	eigen	100/100	5.17 ± 1.039
34	custom	gumbel	10/10	5.271 ± 1.41
35	pfjax	ot	10/10	5.328 ± 1.421
36	filterflow	corrected_ot	10/10	5.335 ± 1.44
37	pydpf	ot	10/10	5.416 ± 1.463
38	filterflow	ot	10/10	5.48 ± 1.493

Table 21: UNGM filter ($N = 128$): 95% coverage. Best = closest to 95% is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	95% coverage
1	pfjax	**mvn**	93/100	$\mathbf{0.9372 \pm 0.02179}$
2	cuthbert	systematic	100/100	0.9348 ± 0.02619
2	cuthbert	systematic_stop	100/100	0.9348 ± 0.02619
2	julia	multinomial	100/100	0.9348 ± 0.02262
5	julia	ssp	100/100	0.9334 ± 0.02652
5	particles	ssp	100/100	0.9334 ± 0.02391
7	custom	gaussian	100/100	0.933 ± 0.02723
8	filterflow	multinomial	100/100	0.9326 ± 0.02342
9	particles	stratified	100/100	0.9322 ± 0.026
9	stonesoup	multinomial	100/100	0.9322 ± 0.02599
11	pydpf	systematic	100/100	0.932 ± 0.02609
11	pydpf	systematic_stop	100/100	0.932 ± 0.02609
13	julia	stratified	100/100	0.9318 ± 0.02484
14	pydpf	multinomial	100/100	0.931 ± 0.02108
14	torchfilter	multinomial	100/100	0.931 ± 0.02108
16	stonesoup	systematic	100/100	0.9308 ± 0.02419
17	filterflow	systematic	100/100	0.9304 ± 0.02363
17	particles	systematic	100/100	0.9304 ± 0.02754
19	filterflow	stratified	100/100	0.9302 ± 0.02318
19	stonesoup	residual	100/100	0.9302 ± 0.02652
21	julia	killing	100/100	0.93 ± 0.02177
21	custom	eigen	100/100	0.93 ± 0.02605
21	variational	kl	100/100	0.93 ± 0.02457
24	pfjax	multinomial	100/100	0.9292 ± 0.02579
25	julia	systematic	100/100	0.9288 ± 0.025
26	particles	multinomial	100/100	0.9284 ± 0.02483
26	variational	tv	100/100	0.9284 ± 0.02409
26	particles	residual	100/100	0.9284 ± 0.02786
29	pydpf	soft	100/100	0.928 ± 0.02138
29	torchfilter	soft	100/100	0.928 ± 0.02138
31	julia	residual	100/100	0.9274 ± 0.0278
32	cuthbert	multinomial	100/100	0.9272 ± 0.02584
33	stonesoup	stratified	100/100	0.925 ± 0.02727
34	custom	gumbel	10/10	0.918 ± 0.04446
34	pfjax	ot	10/10	0.918 ± 0.04182
36	pydpf	ot	10/10	0.914 ± 0.04577
36	filterflow	corrected_ot	10/10	0.914 ± 0.04268
38	filterflow	ot	10/10	0.908 ± 0.04484

Table 22: UNGM filter ($N = 128$): conditional log-likelihood variance. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	conditional log-likelihood variance
1	pydpf	**soft**	100/100	**438.4**
1	torchfilter	**soft**	100/100	**438.4**
3	pfjax	mvn	93/100	529.8
4	particles	stratified	100/100	617.1
5	particles	ssp	100/100	633.7
6	filterflow	systematic	100/100	635.8
7	particles	systematic	100/100	653.3
8	particles	residual	100/100	662.5
9	pydpf	systematic	100/100	669.5
9	pydpf	systematic_stop	100/100	669.5
11	filterflow	stratified	100/100	672.7
12	julia	ssp	100/100	685.2
13	julia	stratified	100/100	719.2
14	filterflow	multinomial	100/100	729.7
15	variational	kl	100/100	756
16	julia	killing	100/100	760.2
17	julia	multinomial	100/100	761.1
18	cuthbert	systematic	100/100	769.7
18	cuthbert	systematic_stop	100/100	769.7
20	custom	eigen	100/100	809.7
21	julia	residual	100/100	836.5
22	stonesoup	residual	100/100	843.9
23	custom	gaussian	100/100	845.1
24	stonesoup	systematic	100/100	953.5
25	stonesoup	multinomial	100/100	969.2
26	pydpf	multinomial	100/100	974.5
26	torchfilter	multinomial	100/100	974.5
28	julia	systematic	100/100	1081
29	cuthbert	multinomial	100/100	1181
30	variational	tv	100/100	2923
31	stonesoup	stratified	100/100	3469
32	particles	multinomial	100/100	4174
33	pfjax	multinomial	100/100	6550
–	custom	gumbel	10/10	–
–	filterflow	corrected_ot	10/10	–
–	filterflow	ot	10/10	–
–	pfjax	ot	10/10	–
–	pydpf	ot	10/10	–

Table 23: Stochastic volatility filter ($N = 128$): RMSE. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	RMSE
1	custom	**gumbel**	10/10	$\mathbf{0.4148 \pm 0.06775}$
2	stonesoup	stratified	100/100	0.4151 ± 0.06275
3	stonesoup	systematic	100/100	0.4153 ± 0.06295
4	filterflow	multinomial	100/100	0.4158 ± 0.06228
5	custom	gaussian	100/100	0.4158 ± 0.06298
6	filterflow	stratified	100/100	0.4163 ± 0.06335
7	pfjax	mvn	100/100	0.4163 ± 0.06235
8	julia	multinomial	100/100	0.4163 ± 0.06232
9	custom	eigen	100/100	0.4164 ± 0.06297
10	particles	ssp	100/100	0.4164 ± 0.06348
11	pydpf	systematic	100/100	0.4165 ± 0.06394
11	pydpf	systematic_stop	100/100	0.4165 ± 0.06394
13	julia	ssp	100/100	0.4165 ± 0.06392
14	pydpf	multinomial	100/100	0.4167 ± 0.06288
14	torchfilter	multinomial	100/100	0.4167 ± 0.06288
16	filterflow	systematic	100/100	0.4168 ± 0.06332
17	cuthbert	systematic	100/100	0.4171 ± 0.06367
17	cuthbert	systematic_stop	100/100	0.4171 ± 0.06367
19	stonesoup	residual	100/100	0.4171 ± 0.06352
20	julia	stratified	100/100	0.4173 ± 0.06348
21	particles	stratified	100/100	0.4173 ± 0.06379
22	particles	systematic	100/100	0.4173 ± 0.0634
23	filterflow	ot	10/10	0.4173 ± 0.065
24	particles	multinomial	100/100	0.4174 ± 0.06486
25	variational	kl	100/100	0.4174 ± 0.06379
26	julia	systematic	100/100	0.4174 ± 0.0627
27	julia	residual	100/100	0.4174 ± 0.06273
28	filterflow	corrected_ot	10/10	0.4176 ± 0.06451
29	cuthbert	multinomial	100/100	0.4176 ± 0.06238
30	pydpf	ot	10/10	0.4181 ± 0.06481
31	variational	tv	100/100	0.4181 ± 0.06438
32	pfjax	ot	10/10	0.4183 ± 0.06485
33	particles	residual	100/100	0.4183 ± 0.06283
34	pfjax	multinomial	100/100	0.4186 ± 0.06349
35	stonesoup	multinomial	100/100	0.4189 ± 0.06357
36	pydpf	soft	100/100	0.419 ± 0.06346
36	torchfilter	soft	100/100	0.419 ± 0.06346
38	julia	killing	100/100	0.4191 ± 0.06314

Table 24: Stochastic volatility filter ($N = 128$): 95% coverage. Best = closest to 95% is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	95% coverage
1	custom	**gumbel**	10/10	$\mathbf{0.946 \pm 0.03697}$
2	variational	tv	100/100	0.9562 ± 0.02419
3	particles	residual	100/100	0.9592 ± 0.02316
4	pydpf	soft	100/100	0.9594 ± 0.02177
4	torchfilter	soft	100/100	0.9594 ± 0.02177
6	stonesoup	multinomial	100/100	0.9598 ± 0.02265
7	variational	kl	100/100	0.9604 ± 0.02065
8	pfjax	multinomial	100/100	0.961 ± 0.02148
9	stonesoup	residual	100/100	0.9612 ± 0.02046
10	particles	multinomial	100/100	0.9614 ± 0.0219
11	cuthbert	systematic	100/100	0.9616 ± 0.02213
11	cuthbert	systematic_stop	100/100	0.9616 ± 0.02213
13	cuthbert	multinomial	100/100	0.9622 ± 0.02241
14	julia	killing	100/100	0.9624 ± 0.02087
15	julia	stratified	100/100	0.9626 ± 0.02047
16	filterflow	stratified	100/100	0.9628 ± 0.01923
17	pydpf	multinomial	100/100	0.9632 ± 0.01949
17	torchfilter	multinomial	100/100	0.9632 ± 0.01949
19	julia	multinomial	100/100	0.9634 ± 0.02093
20	filterflow	systematic	100/100	0.9636 ± 0.0214
21	julia	ssp	100/100	0.9644 ± 0.01968
21	stonesoup	stratified	100/100	0.9644 ± 0.0207
23	custom	gaussian	100/100	0.9646 ± 0.0197
24	filterflow	multinomial	100/100	0.9648 ± 0.01861
24	particles	stratified	100/100	0.9648 ± 0.02066
26	julia	residual	100/100	0.965 ± 0.01826
26	stonesoup	systematic	100/100	0.965 ± 0.02031
28	particles	ssp	100/100	0.9656 ± 0.02095
29	particles	systematic	100/100	0.9658 ± 0.01838
30	filterflow	corrected_ot	10/10	0.966 ± 0.02138
31	custom	eigen	100/100	0.9662 ± 0.02032
32	pfjax	mvn	100/100	0.9664 ± 0.01985
33	julia	systematic	100/100	0.9668 ± 0.01836
34	pydpf	systematic	100/100	0.967 ± 0.01967
34	pydpf	systematic_stop	100/100	0.967 ± 0.01967
36	filterflow	ot	10/10	0.968 ± 0.02046
37	pfjax	ot	10/10	0.97 ± 0.01937
37	pydpf	ot	10/10	0.97 ± 0.01937

Table 25: Stochastic volatility filter ($N = 128$): conditional log-likelihood variance. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	conditional log-likelihood variance
1	stonesoup	**stratified**	100/100	**0.07439**
2	custom	eigen	100/100	0.07457
3	julia	multinomial	100/100	0.07481
4	cuthbert	systematic	100/100	0.07696
4	cuthbert	systematic_stop	100/100	0.07696
6	custom	gaussian	100/100	0.07859
7	stonesoup	systematic	100/100	0.07898
8	julia	ssp	100/100	0.07956
9	pydpf	systematic	100/100	0.08059
9	pydpf	systematic_stop	100/100	0.08059
11	stonesoup	residual	100/100	0.08088
12	particles	systematic	100/100	0.08285
13	julia	systematic	100/100	0.08305
14	particles	ssp	100/100	0.08356
15	pydpf	multinomial	100/100	0.08358
15	torchfilter	multinomial	100/100	0.08358
17	variational	tv	100/100	0.08647
18	julia	stratified	100/100	0.08726
19	pfjax	mvn	100/100	0.08799
20	julia	residual	100/100	0.08983
21	particles	residual	100/100	0.09107
22	filterflow	stratified	100/100	0.09186
23	pfjax	multinomial	100/100	0.09211
24	variational	kl	100/100	0.09313
25	particles	stratified	100/100	0.0933
26	julia	killing	100/100	0.09429
27	filterflow	multinomial	100/100	0.09982
28	filterflow	systematic	100/100	0.1003
29	stonesoup	multinomial	100/100	0.1031
30	cuthbert	multinomial	100/100	0.1066
31	particles	multinomial	100/100	0.107
32	pydpf	soft	100/100	0.1155
32	torchfilter	soft	100/100	0.1155
–	custom	gumbel	10/10	–
–	filterflow	corrected_ot	10/10	–
–	filterflow	ot	10/10	–
–	pfjax	ot	10/10	–
–	pydpf	ot	10/10	–

Table 26: Bearing-only tracking filter ($N = 128$): RMSE. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	RMSE
1	filterflow	**corrected_ot**	10/10	$\mathbf{3.25 \pm 1.011}$
2	julia	killing	100/100	4.442 ± 1.488
3	julia	ssp	100/100	4.478 ± 1.728
4	filterflow	systematic	100/100	4.487 ± 1.609
5	custom	gaussian	100/100	4.508 ± 1.737
6	custom	eigen	100/100	4.51 ± 1.71
7	julia	systematic	100/100	4.529 ± 1.732
8	particles	systematic	100/100	4.548 ± 1.657
9	cuthbert	systematic	100/100	4.553 ± 1.764
9	cuthbert	systematic_stop	100/100	4.553 ± 1.764
11	julia	stratified	100/100	4.561 ± 1.735
12	stonesoup	systematic	100/100	4.6 ± 1.616
13	particles	ssp	100/100	4.649 ± 1.688
14	stonesoup	multinomial	100/100	4.649 ± 1.548
15	variational	kl	100/100	4.662 ± 1.626
16	stonesoup	stratified	100/100	4.698 ± 1.65
17	filterflow	stratified	100/100	4.705 ± 1.677
18	pfjax	mvn	100/100	4.721 ± 1.71
19	stonesoup	residual	100/100	4.733 ± 1.612
20	particles	residual	100/100	4.746 ± 1.683
21	pydpf	systematic	100/100	4.763 ± 1.746
21	pydpf	systematic_stop	100/100	4.763 ± 1.746
23	julia	residual	100/100	4.763 ± 1.596
24	cuthbert	multinomial	100/100	4.773 ± 1.757
25	particles	stratified	100/100	4.775 ± 1.675
26	pydpf	multinomial	100/100	4.776 ± 1.538
26	torchfilter	multinomial	100/100	4.776 ± 1.538
28	pfjax	multinomial	100/100	4.777 ± 1.758
29	particles	multinomial	100/100	4.787 ± 1.635
30	filterflow	multinomial	100/100	4.791 ± 1.673
31	torchfilter	soft	100/100	4.799 ± 1.711
31	pydpf	soft	100/100	4.799 ± 1.711
33	julia	multinomial	100/100	4.807 ± 1.856
34	variational	tv	100/100	4.869 ± 1.788
35	custom	gumbel	10/10	5.188 ± 2.52
36	filterflow	ot	10/10	5.296 ± 1.728
37	pfjax	ot	10/10	5.559 ± 2.289
38	pydpf	ot	10/10	5.648 ± 2.284

Table 27: Bearing-only tracking filter ($N = 128$): 95% coverage. Best = closest to 95% is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	95% coverage
1	custom	**eigen**	100/100	$\mathbf{0.7776 \pm 0.1978}$
2	pfjax	mvn	100/100	0.7414 ± 0.1762
3	custom	gaussian	100/100	0.724 ± 0.1915
4	filterflow	corrected_ot	10/10	0.72 ± 0.1564
5	julia	ssp	100/100	0.6614 ± 0.2063
6	julia	stratified	100/100	0.6586 ± 0.1988
7	julia	systematic	100/100	0.6522 ± 0.2128
8	filterflow	systematic	100/100	0.6508 ± 0.2106
9	pydpf	systematic	100/100	0.6492 ± 0.1938
9	pydpf	systematic_stop	100/100	0.6492 ± 0.1938
11	cuthbert	systematic	100/100	0.6478 ± 0.2098
11	cuthbert	systematic_stop	100/100	0.6478 ± 0.2098
13	particles	systematic	100/100	0.6364 ± 0.2056
14	particles	ssp	100/100	0.633 ± 0.2233
15	variational	kl	100/100	0.6326 ± 0.1913
16	stonesoup	systematic	100/100	0.6286 ± 0.2024
17	julia	killing	100/100	0.6212 ± 0.1858
18	particles	residual	100/100	0.62 ± 0.2207
19	filterflow	stratified	100/100	0.6176 ± 0.2069
20	particles	stratified	100/100	0.6112 ± 0.1851
21	stonesoup	residual	100/100	0.6078 ± 0.2107
22	stonesoup	stratified	100/100	0.6044 ± 0.1925
23	julia	residual	100/100	0.6006 ± 0.2023
24	variational	tv	100/100	0.5896 ± 0.1898
25	cuthbert	multinomial	100/100	0.5798 ± 0.2044
26	pfjax	multinomial	100/100	0.5742 ± 0.2113
27	pydpf	multinomial	100/100	0.5696 ± 0.2071
27	torchfilter	multinomial	100/100	0.5696 ± 0.2071
29	stonesoup	multinomial	100/100	0.5648 ± 0.1863
30	filterflow	multinomial	100/100	0.56 ± 0.1744
31	julia	multinomial	100/100	0.5588 ± 0.2031
32	particles	multinomial	100/100	0.5396 ± 0.2098
33	pydpf	soft	100/100	0.5192 ± 0.1963
33	torchfilter	soft	100/100	0.5192 ± 0.1963
35	custom	gumbel	10/10	0.216 ± 0.1761
36	filterflow	ot	10/10	0.052 ± 0.06546
37	pfjax	ot	10/10	0.042 ± 0.02564
38	pydpf	ot	10/10	0.04 ± 0.02612

Table 28: Bearing-only tracking filter ($N = 128$): conditional log-likelihood variance. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	conditional log-likelihood variance
1	custom	**eigen**	100/100	**0.586**
2	pfjax	mvn	100/100	1.197
3	custom	gaussian	100/100	1.992
4	particles	ssp	100/100	2.019
5	julia	multinomial	100/100	2.921
6	stonesoup	stratified	100/100	2.93
7	particles	stratified	100/100	3.091
8	cuthbert	systematic	100/100	3.326
8	cuthbert	systematic_stop	100/100	3.326
10	stonesoup	systematic	100/100	3.771
11	julia	stratified	100/100	3.849
12	filterflow	multinomial	100/100	3.972
13	pfjax	multinomial	100/100	4.521
14	pydpf	multinomial	100/100	4.556
14	torchfilter	multinomial	100/100	4.556
16	julia	ssp	100/100	4.573
17	pydpf	systematic	100/100	4.651
17	pydpf	systematic_stop	100/100	4.651
19	julia	killing	100/100	4.723
20	particles	systematic	100/100	4.739
21	stonesoup	multinomial	100/100	4.879
22	julia	systematic	100/100	5.275
23	variational	tv	100/100	6.066
24	cuthbert	multinomial	100/100	6.607
25	filterflow	stratified	100/100	7.085
26	filterflow	systematic	100/100	7.311
27	variational	kl	100/100	7.522
28	particles	residual	100/100	7.577
29	julia	residual	100/100	7.695
30	stonesoup	residual	100/100	7.747
31	pydpf	soft	100/100	9.433
31	torchfilter	soft	100/100	9.433
33	particles	multinomial	100/100	19.1
–	custom	gumbel	10/10	–
–	filterflow	corrected_ot	10/10	–
–	filterflow	ot	10/10	–
–	pfjax	ot	10/10	–
–	pydpf	ot	10/10	–

Table 29: PF likelihood gradient vs exact Kalman ($N = 128$): gradient bias. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	gradient bias
1	custom	**eigen**	100	**0.5272**
2	custom	gaussian	100	0.8231
3	pydpf	systematic_stop	100	1.748
4	cuthbert	systematic_stop	100	2.285
5	pydpf	soft	100	6.675
5	torchfilter	soft	100	6.675
7	custom	gumbel	10	8.148
8	pydpf	systematic	100	11.27
9	cuthbert	systematic	100	12.98
10	pydpf	ot	10	14.21

Table 30: PF likelihood gradient vs exact Kalman ($N = 128$): gradient variance. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	gradient variance
1	custom	**eigen**	100	**5.207**
2	custom	gaussian	100	5.602
3	custom	gumbel	10	20.28
4	pydpf	systematic	100	41.82
5	cuthbert	systematic	100	44.41
6	pydpf	soft	100	51.45
6	torchfilter	soft	100	51.45
8	pydpf	systematic_stop	100	150.6
9	cuthbert	systematic_stop	100	175.9
10	pydpf	ot	10	714.6

Table 31: PF likelihood gradient vs exact Kalman ($N = 128$): gradient MSE. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	gradient MSE
1	custom	**eigen**	100	**5.433**
2	custom	gaussian	100	6.223
3	custom	gumbel	10	84.64
4	pydpf	soft	100	95.5
4	torchfilter	soft	100	95.5
6	pydpf	systematic_stop	100	152.2
7	pydpf	systematic	100	168.4
8	cuthbert	systematic_stop	100	179.4
9	cuthbert	systematic	100	212.5
10	pydpf	ot	10	844.9

Table 32: PF likelihood gradient vs exact Kalman ($N = 128$): log-likelihood variance. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	log-likelihood variance
1	pydpf	**ot**	10	**0.3098**
2	custom	eigen	100	0.5643
3	cuthbert	systematic	100	0.621
3	cuthbert	systematic_stop	100	0.621
5	pydpf	systematic_stop	100	0.6556
5	pydpf	systematic	100	0.6556
7	custom	gumbel	10	0.7297
8	custom	gaussian	100	0.86
9	pydpf	soft	100	1.081
9	torchfilter	soft	100	1.081

Table 33: PF likelihood gradient vs exact Kalman ($N = 128$): log-likelihood bias. Best = closest to 0 is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	log-likelihood bias
1	custom	**eigen**	100	**−0.1536**
2	cuthbert	systematic	100	−0.2671
2	cuthbert	systematic_stop	100	−0.2671
4	custom	gaussian	100	−0.2781
5	pydpf	systematic	100	−0.2953
5	pydpf	systematic_stop	100	−0.2953
7	pydpf	soft	100	−0.3859
7	torchfilter	soft	100	−0.3859
9	pydpf	ot	10	−0.4472
10	custom	gumbel	10	−2.662

Table 34: SV observation-scale learning ($N = 128$): $|\hat{s} - 1|$. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	$\lvert\hat{s} - 1\rvert$
1	torchfilter	**soft**	10/10	$\mathbf{0.2 \pm 0.09538}$
2	custom	gaussian	10/10	0.22 ± 0.1056
2	cuthbert	systematic_stop	10/10	0.22 ± 0.1253
2	particles	systematic	10/10	0.22 ± 0.1253
5	stonesoup	systematic	10/10	0.24 ± 0.1129
5	variational	kl	10/10	0.24 ± 0.1129
7	custom	eigen	10/10	0.26 ± 0.09657
7	custom	gumbel	10/10	0.26 ± 0.09657
7	pfjax	ot	10/10	0.26 ± 0.09657
7	pydpf	ot	10/10	0.26 ± 0.09657
11	filterflow	ot	10/10	0.28 ± 0.1
11	julia	systematic	10/10	0.28 ± 0.07388

Table 35: SV observation-scale learning ($N = 128$): $|\hat{s} - \hat{s}_{\text{ref}}|$. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	$\lvert\hat{s} - \hat{s}_{\text{ref}}\rvert$
1	cuthbert	**systematic_stop**	10/10	$\mathbf{0.02 \pm 0.04524}$
1	particles	**systematic**	10/10	$\mathbf{0.02 \pm 0.04524}$
3	variational	kl	10/10	0.04 ± 0.06032
4	custom	eigen	10/10	0.06 ± 0.06911
4	custom	gaussian	10/10	0.06 ± 0.06911
4	pfjax	ot	10/10	0.06 ± 0.06911
4	pydpf	ot	10/10	0.06 ± 0.06911
8	filterflow	ot	10/10	0.08 ± 0.07388
8	stonesoup	systematic	10/10	0.08 ± 0.07388
8	torchfilter	soft	10/10	0.08 ± 0.07388
11	custom	gumbel	10/10	0.1 ± 0.07541
12	julia	systematic	10/10	0.12 ± 0.07388

Table 36: SV observation-scale learning ($N = 128$): likelihood-score RMSE. Best = lower is better. Best method in bold; ± gives the 95% confidence-interval half-width where available.

Rank	Provider	Method	Trials	likelihood-score RMSE
1	pydpf	**ot**	10/10	**1.146**
2	pfjax	ot	10/10	1.179
3	filterflow	ot	10/10	1.307
4	custom	eigen	10/10	2.112
5	custom	gumbel	10/10	4.606
6	custom	gaussian	10/10	5.252
7	variational	kl	10/10	6.566
8	julia	systematic	10/10	7.394
9	cuthbert	systematic_stop	10/10	7.468
10	particles	systematic	10/10	8.793
11	torchfilter	soft	10/10	10.14
12	stonesoup	systematic	10/10	10.19